4 Power Question 2016: Taxicab Geometry

Instructions: The power question is worth 50 points; each part's point value is given in brackets next to the part. To receive full credit, the presentation must be legible, orderly, clear, and concise. If a problem says "list" or "compute," you need not justify your answer. If a problem says "determine," "find," or "show," then you must show your work or explain your reasoning to receive full credit, although such explanations do not have to be lengthy. If a problem says "justify" or "prove," then you must prove your answer rigorously. Even if not proved, earlier numbered items may be used in solutions to later numbered items, but not vice versa. Pages submitted for credit should be NUMBERED IN CONSECUTIVE ORDER AT THE TOP OF EACH PAGE in what your team considers to be proper sequential order. PLEASE WRITE ON ONLY ONE SIDE OF THE ANSWER PAPERS. Put the TEAM NUMBER (not the team name) on the cover sheet used as the first page of the papers submitted. Do not identify the team in any other way.

In standard coordinate (Euclidean) geometry, if $A = (x_1, y_1)$ and $B = (x_2, y_2)$, then the distance between A and B is given by the Euclidean distance function: $d_E(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. In this Power Question, we investigate what happens when distance is measured using the *taxicab* distance function,

$$d_T(A, B) = |x_2 - x_1| + |y_2 - y_1|.$$

This distance is called the *taxi-distance* between A and B.

1. This problem will help you become familiar with taxi-distances.

- **a**. Compute the taxi-distance between the points $(2, 6)$ and $(7, -2)$. [1 pt]
- **b**. Compute the coordinates of a point whose Euclidean distance from $(3, 2)$ is 5 and whose taxi-distance from $(3, 2)$ is also 5. [1 pt]
- **c**. Compute the coordinates of a point whose Euclidean distance from $(3, 2)$ is 5 and whose taxi-distance from $(3, 2)$ is 7. [1 pt]

In Euclidean geometry, one possible definition of the line segment $\overline{AB}$ is the set of points P such that $AP+PB = AB$, or $d_E(A, P) + d_E(P, B) = d_E(A, B)$. This definition does not work as well in taxicab geometry!

2. Let $A = (0, 4)$ and $B = (4, 0)$.

- **a**. Show that $P = (1, 3)$ satisfies $d_T(A, P) + d_T(P, B) = d_T(A, B)$. [1 pt]
- **b**. Find the coordinates of a point P *not* on the line $x + y = 4$ satisfying $d_T(A, P) + d_T(P, B) = d_T(A, B)$. [1 pt]
- **c**. Find the set of all points P such that $d_T(A, P) + d_T(P, B) = d_T(A, B)$, and justify your answer. [3 pts]

As in Euclidean geometry, two triangles are congruent if and only if the lengths of corresponding sides and the measures of corresponding angles are equal. In Euclidean geometry, this *measure-congruence* is equivalent to *transformation-congruence*: two triangles are transformation-congruent if and only if one is the image of the other under a composition of reflections, rotations, and translations. (Actually, reflections alone suffice, but that result is beyond the scope of this Power Question.) For the remainder of this Power Question, the symbol $\cong$ will be used exclusively to denote measure-congruence; we write $\cong_T$ to emphasize that we are using taxicab measure.

3. Let's investigate whether the equivalence between measure- and transformation-congruence holds in taxicab geometry. Suppose that $A = (x_A, y_A)$ and similarly for other named points.

- **a**. If $\triangle A'B'C'$ is the image of $\triangle ABC$ under the translation $(x, y) \mapsto (x + h, y + k)$, are the taxi-lengths of the sides of $\triangle A'B'C'$ equal to those of $\triangle ABC$? Justify your answer. [1 pt]
- **b**. If $\triangle A'B'C'$ is the image of $\triangle ABC$ under a reflection in the line $y = k$, are the taxi-lengths of the sides of $\triangle A'B'C'$ equal to those of $\triangle ABC$? Justify your answer. [1 pt]
- **c**. Let $O = (0, 0)$, $R = (8, 0)$, $D = (0, 4)$, and let $\triangle O'R'D'$ be the image of $\triangle ORD$ under a (Euclidean) rotation of 45° counterclockwise about the origin. Is $\triangle ORD \cong_T \triangle O'R'D'$? Justify your answer. [2 pts]

4. Let O have coordinates $(0,0)$, let P have coordinates $(1,0)$, and let A be the point $(4,2)$. The point B has coordinates (m,n), where m and n are both positive integers, m is relatively prime to n, and $\angle BOA \cong \angle AOP$ (considered as Euclidean angles).

a. Compute the ordered pair (m,n). [2 pts]

b. Compute the taxi-lengths of the segments intercepted by $\angle BOA$ and $\angle AOP$ on the line $x+y=1$. [2 pts]

The computation in the previous item is problematic, for the following reason. A *taxi-circle* is the set of all points that are a fixed taxi-distance from a given point, and the segments whose lengths you computed in 4b are actually arcs on a unit taxi-circle centered at the origin. So angles congruent in the Euclidean sense do not necessarily intercept taxi-congruent arcs.

One way to fix this problem is to define the measure of an angle analogously to the way the radian measure of an angle is defined in Euclidean geometry, namely, as a quotient. The numerator is the length of the arc that the angle intercepts on a circle centered at the angle's vertex. The denominator is the radius of that circle. If P and R are equidistant from Q, then the *measure of $\angle PQR$ in taxi-radians* is given by the formula

$$\mathrm{m}_T\angle PQR = \frac{\text{taxi-length of the arc from } P \text{ to } R \text{ on the taxi-circle centered at } Q \text{ passing through } P}{d_T(Q,P)}.$$

5. Let's flesh out the above definition with an example.

a. Find an equation to describe the taxi-circle centered at the origin of radius 17. [1 pt]

b. Compute the value of *taxi-pi*, π_T, by dividing the semi-taxi-circumference of the taxi-circle from part 5a by its radius. [1 pt]

c. Show that the value you computed in part 5b is independent of the taxi-circle's radius. [2 pts]

d. Given that $C=(3,1)$, $O=(0,0)$, and $B=(-1,3)$, compute $\mathrm{m}_T\angle COB$. [1 pt]

Using taxi-circles and taxi-angles, define a *rotation by θ taxi-radians about the origin* as follows. The point P' is the image of P under a counterclockwise rotation of θ taxi-radians around $O=(0,0)$ if and only if $d_T(O,P)=d_T(O,P')$ and the measure of the *taxi-arc* from P to P' (going counterclockwise around the taxi-circle centered at O) is $d_T(O,P)\cdot\theta$.

6. Let $O=(0,0)$, $A=(8,0)$, and $R=(3,1)$.

a. Let $\triangle O'A'R'$ be the image of $\triangle OAR$ under a rotation of 1 taxi-radian about the origin. Compute the coordinates of O', A', and R'. [3 pts]

b. Is $\triangle O'A'R' \cong_T \triangle OAR$? Justify your answer. [3 pts]

7. We now consider conditions for two triangles to be measure-congruent in Euclidean geometry and apply them to taxicab geometry to discover whether or not measure-congruence still holds. Let $\triangle MOD$ have vertices $M=(0,2)$, $O=(0,0)$, and $D=(2,0)$.

a. Show that there exists a triangle $\triangle CAB$ such that $d_T(C,A)=d_T(M,O)$, $d_T(A,B)=d_T(O,D)$, and $\mathrm{m}_T\angle CAB = \mathrm{m}_T\angle MOD$, but $\triangle MOD \ncong_T \triangle CAB$. (Thus SAS congruence does not hold in taxicab geometry.) [2 pts]

b. Show that there exists a triangle $\triangle PIG$ such that $d_T(P,I)=d_T(M,O)$, $d_T(I,G)=d_T(O,D)$, and $d_T(P,G)=d_T(M,D)$, but $\triangle MOD \ncong_T \triangle PIG$. (Thus SSS congruence does not hold in taxicab geometry.) [2 pts]

8. Define the *taxi-cosine* and *taxi-sine* of an angle as follows:

$$\begin{aligned}\cos_T\angle ACB &= \frac{d_E(C,A)\cdot d_E(C,B)}{d_T(C,A)\cdot d_T(C,B)}\cos\angle ACB\\ \sin_T\angle ACB &= \frac{d_E(C,A)\cdot d_E(C,B)}{d_T(C,A)\cdot d_T(C,B)}\sin\angle ACB,\end{aligned}$$

where $\cos\angle ACB$ and $\sin\angle ACB$ are the ordinary (Euclidean) cosine and sine functions, respectively.

a. Given $A = (3, 0)$, $O = (0, 0)$, and $B = (3, 4)$, compute $\cos_T \angle AOB$. [1 pt]

b. Show that if $\overrightarrow{OA}$ is parallel to the x-axis, then $\cos_T \angle AOB = \dfrac{\cos \angle AOB}{|\cos \angle AOB| + |\sin \angle AOB|}$. [3 pts]

c. Let $\overrightarrow{QP}$ be parallel to the x-axis, and let A and B be points not on $\overrightarrow{QP}$. Show that

$$\cos_T \angle AQB = \frac{\cos \angle AQB}{(|\cos \angle AQP| + |\sin \angle AQP|)(|\cos \angle BQP| + |\sin \angle BQP|)}.$$

[3 pts]

9. Suppose that triangle ABC is given with $\overline{AB}$ parallel to the x-axis, and that the perpendicular dropped from C to $\overleftrightarrow{AB}$ intersects $\overline{AB}$ at C' and has taxi-length h_T. As with conventional trigonometry, let $a_T = d_T(B, C)$, $b_T = d_T(A, C)$, and $c_T = d_T(A, B)$. Prove the following:

a. $a_T = b_T + c_T - 2b_T \cos_T A$. [3 pts]

b. $b_T = a_T + c_T - 2a_T \cos_T B$. [2 pts]

c. $c_T = \dfrac{a_T^2 + b_T^2 - 2a_T b_T \cos_T C}{a_T + b_T}$. [3 pts]

10. Given $\triangle ABC$, prove that $\dfrac{a_T}{\sin_T A} = \dfrac{b_T}{\sin_T B} = \dfrac{c_T}{\sin_T C}$, where a_T, b_T, and c_T are defined as in the previous problem. [4 pts]